

# Maestro — TUI Design

Wireframes, interaction flows and settings inventory

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Companion to the design decisions (FR-10). All wireframes are schematic mock-ups of AppCUI windows/controls (rendered as vector figures in the PDF build). AppCUI model: a **desktop** hosts multiple **windows**; a bottom/top **app bar** shows global status; every action is reachable by keyboard, menu and mouse.

## 1. Design Principles

- **One Mission Control, many satellites.** The main window (tabs) is for global state; every *project* and every *session* can open as its own satellite window so parallel work is visible side by side on the desktop.
- **Live by default.** Every list/gauge subscribes to the event bus; no refresh buttons.
- **Nothing hidden.** Pending questions, write-queue depth and quota alarms are always one glance away (app bar + dashboard ticker).
- **Keyboard-first.** Global hotkeys work from any window; every form is tab-navigable.
- **Same core for TUI and CLI** — the TUI is a view; `maestro-cli` can do everything.

## 2. Global Chrome

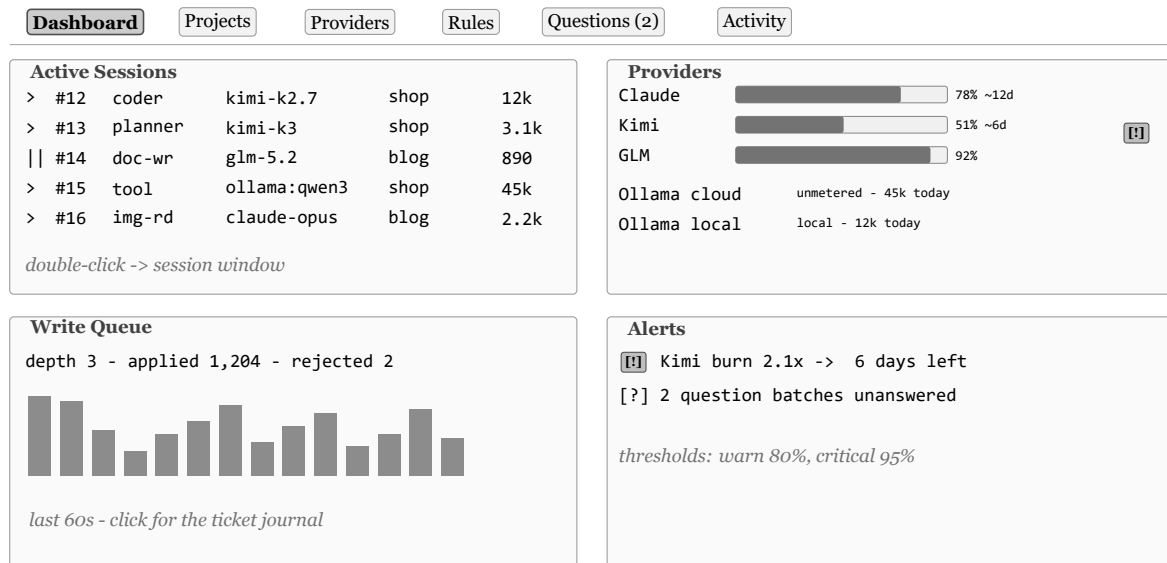


- **App bar segments** (left->right): active sessions • tokens today • cost today • write-queue depth • pending clarifications (click -> Interrogation Center) • active alerts (click -> alerts log) • clock. Segments are clickable buttons.
- **Menu map:**
  - *File*: New Project, Open Workspace, Import/Export Config, Quit
  - *Project*: Run/Pause/Resume, New Task, Interview, Regenerate AGENTS.md, Skills...
  - *Providers*: Add Provider, Detect CLIs, Health Check All, Quota Report
  - *Rules*: New Rule, Dry-Run, Import YAML
  - *Sessions*: Quick Chat, New Task Session, Pause All, Transcripts...
  - *View*: Mission Control, Tile Windows, Theme, Focus Mode
  - *Tools*: Settings, Config Doctor, Logs, Sandbox Allow-Lists
  - *Help*: Shortcuts, About

## 3. Mission Control (main window, opens maximized)

Tab control with 6 fixed tabs: **Dashboard** • **Projects** • **Providers** • **Rules** • **Questions** • **Activity**.

### 3.1 Dashboard tab



- Sessions listview: double-click -> session window; right-click -> pause/steer/kill/migrate.
- Provider gauges: progress bars + projected-exhaustion estimate; click -> provider window.
- Write queue: depth + applied/rejected counters + sparkline (GraphView); click -> ticket log.

### 3.2 Projects tab

| Projects      |           |          |           |        |           |     |
|---------------|-----------|----------|-----------|--------|-----------|-----|
| name          | state     | tasks    | tok today | cost   | questions |     |
| shop-redesign | > running | 2/6 done | 60.3k     | \$1.42 | 2 pending | [!] |
| blog-engine   | paused    | 1/4 done | 3.1k      | \$0.41 | -         |     |
| scraper       | [ok] done | 5/5 done | -         | \$2.07 | -         |     |

[Enter] open [Ctrl+N] new [Del] archive [I] interview

### 3.3 Providers tab

| name         | type          | status | models | quota        | latency | cost today |
|--------------|---------------|--------|--------|--------------|---------|------------|
| Claude       | api           | (*) ok | 4      | 78% left     | 380ms   | \$0.92     |
| Kimi         | api           | (*) ok | 3      | 51% left [!] | 520ms   | \$0.44     |
| claude       | cli           | (*) ok | -      | parsed: 61%  | -       | -          |
| ollama-local | ollama-local  | (*) ok | 20     | unmetered    | 90ms    | \$0.00     |
| ollama-vc    | ollama-remote | (*) ok | 14     | ssh tunnel   | 9ms     | \$0.00     |

[+ Add] [Detect CLIs] [Health Check] Enter -> provider detail (capability matrix, quota, tunnel)

**Provider window** (double-click): tabs *Details* · *Models* · *Quota* · *Adapter*.

- Details: type, endpoint/CLI path, key reference (never the secret — [••• stored in keychain], [Change] [Test] buttons).
- Models: capability matrix (context, vision, tools, streaming, \$/M tok) — editable grid.
- Quota: plan limit, cycle, renewal date, source picker (api/parsed/manual/estimated), burn graph, alert thresholds.
- Adapter (CLI type only): invocation template, output parser, quota parser, overlay mode.

### 3.4 Rules tab

**rules**

```
(*) coder -> kimi / cloud / local
( ) planner -> kimi-k3 -> claude
( ) summarizer -> kimi-k3 / opus
( ) tool-runner -> ollama local
( ) img-reader -> mistral-small
```

*one rule per role + scope;  
priority decides evaluation order*

**editor**

role:

when:  (kimi)

then: 1.   
2.   
3.

scope: (\*) all projects ( ) shop

coder right now -> kimi/kimi-for-coding  
skipped: none - kimi quota 51% >= min 20%

- Left: rule list (one per role+scope); right: condition builder (comboboxes + numeric fields) and **ordered fallback chain** editor.
- **Dry-run panel** evaluates the rule against live quota state and explains the choice.
- "YAML =>" toggle shows the canonical YAML (decision D2) with live two-way sync.

### 3.5 Questions tab — Interrogation Center

**pending batches**

```
> shop-redesign - auth (4q)
  blocks tasks #4, #7

blog-engine - seo (2q)
  advisory - nothing blocked
```

*answering unblocks paused tasks  
automatically (FR-6.3)*

**batch: shop-redesign - auth**

1. Session store?  
(\*) JWT ( ) cookies ( ) other

2. Password reset via email?  
[x] yes -> provider

3. Rate-limit login attempts?  
( ) no (\*) 5/min per IP

*answers land in PRD.json clarifications[] with timestamps*

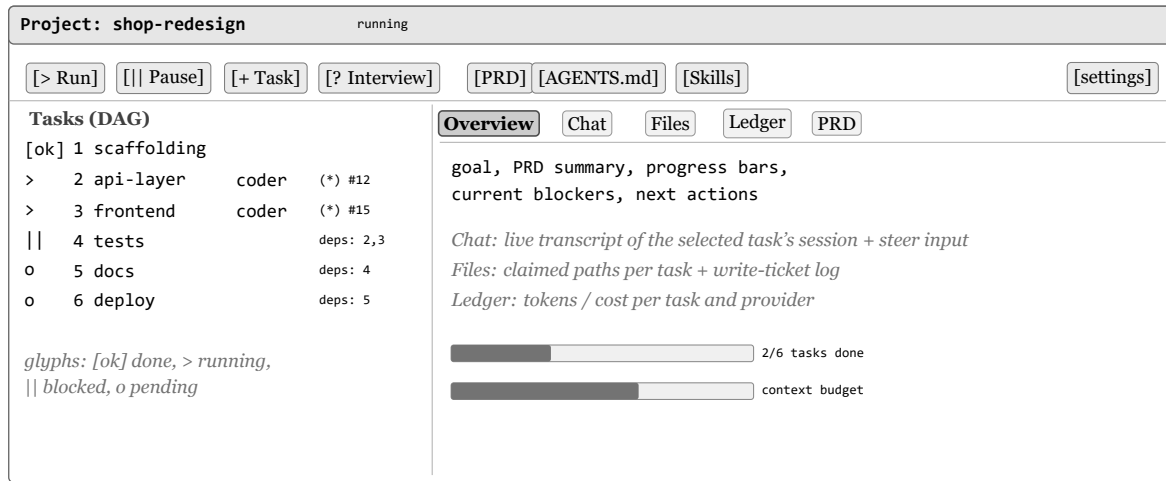
Answering unblocks paused tasks automatically (FR-6.3). "Later" keeps tasks paused but never loses the batch.

### 3.6 Activity tab

Chronological ledger stream (filterable by project/provider/type): tickets applied, rule decisions, quota events, migrations, alerts. Export CSV/JSON buttons.

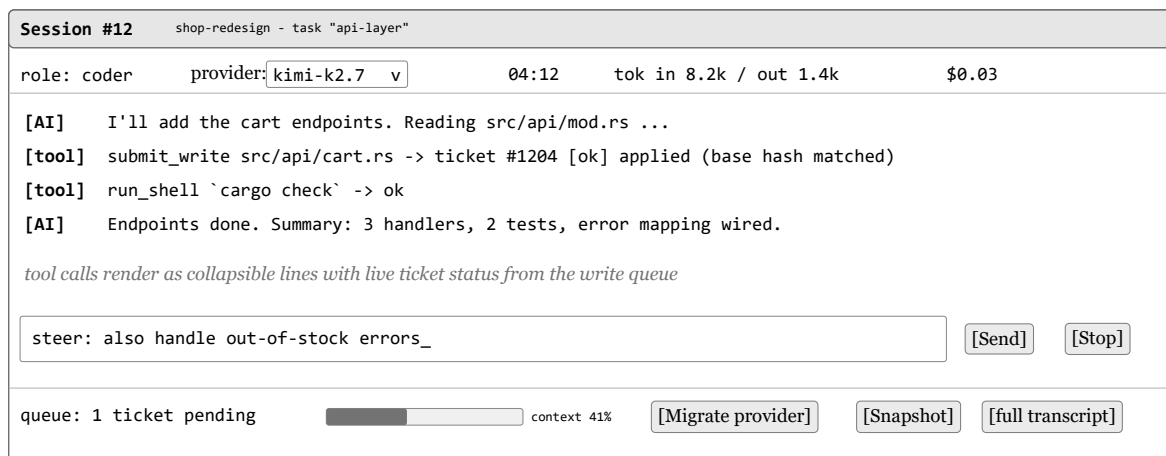
## 4. Project Window (satellite, one per project)

Opened from Projects tab (Enter) or `maestro project open shop`.



- Left: **TreeView of the DAG** — state glyphs ([ok] done, > running, || paused/blocked, o pending), role, live session id. Selecting a task binds the right-side tabs.
- Toolbar [?] **Interview** opens the clarification form for this project; **AGENTS.md** previews the generated file with diff on regeneration; **Skills** opens the suggestion/install panel.
- Right-side tabs (splitter draggable): Overview / Chat / Files / Ledger / (+PRD view).

#### 4.1 Chat / Session view (the heart of single-model interaction)



- Transcript: RichTextField, streamed; tool calls rendered as collapsible lines with ticket status pulled live from the write queue.
- Provider combobox: switch model mid-session -> triggers structured-snapshot migration (decision D5), with confirmation.
- Bottom bar: queue state for this session's tickets, context-window usage, actions (Migrate / Snapshot / open full transcript in markdown viewer).

## 5. Quick Chat (single LLM, no project)

Ctrl+Q or Sessions -> Quick Chat. A lightweight window for ad-hoc conversation:

Quick Chat

provider: anthropic v

model: claude-haiku-4-5 v

78%

you summarise the write-queue design in 2 lines

[AI] Agents submit tickets; the orchestrator applies them per path in FIFO order with base-hash checks.

usage is ledgered against the provider - quotas stay honest

-

[Send]

[Close]

tok 1.2k - \$0.01 - session recorded in the ledger

- Token usage is still ledgered against the provider (quotas stay honest).
- **Promote to task:** converts the chat into a project task with the transcript as context seed. Attach ([clip]) adds files/images for vision-capable models.

## 6. Settings

Menu Tools -> Settings (F10). Modal window: left Accordion/ListBox of sections, right side shows the section's form. **[Save]** applies to TOML live; **[Defaults]** resets.

### Settings inventory

| Section                       | Keys                                                                                                                                  |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>General</b>                | workspace root, default project template, language/locale, autostart behavior, check for updates                                      |
| <b>Appearance</b>             | theme (dark/light/custom), true-color toggle, font-size hint, dashboard density, confirm-on-exit                                      |
| <b>Parallelism</b>            | global session cap (default: CPU heuristic, D6), per-provider concurrency limits, write-queue batch size, per-project fairness quota  |
| <b>Quotas &amp; alerts</b>    | warn thresholds (default 80/95%), alert action (notify / auto-switch rule / pause role), burn-rate window, renewal reminders          |
| <b>Rules</b>                  | default question mode (thorough/balanced/minimal — FR-6.4), unmatched-role behavior (ask / cheapest / fail), rule conflict precedence |
| <b>Secrets</b>                | backend: keychain <-> encrypted vault, vault path, master-password timeout, env-var passthrough allow-list                            |
| <b>Sandbox &amp; security</b> | shell allow-list per project, network access for tool-runner, overlay mode for CLI adapters (D3), redaction level                     |
| <b>Projects</b>               | default DAG parallelism, auto git checkpoints on/off, AGENTS.md regeneration policy (ask/auto), skill registry paths                  |
| <b>Storage</b>                | config dir, SQLite path, transcript retention days, ledger export format, backup on exit                                              |
| <b>Notifications</b>          | TUI popups on/off, OS toast on/off, sound, quiet hours                                                                                |
| <b>Keybindings</b>            | full remapping table (searchable)                                                                                                     |
| <b>Headless/API</b>           | default --json, NDJSON watch toggle, exit-code verbosity                                                                              |
| <b>Advanced</b>               | log level, event-bus buffer size, experimental flags, reset all state                                                                 |

## 7. Interaction Flows

### 7.1 First run (onboarding wizard, runs once)

- Welcome -> pick workspace root & theme.
- **Detect providers:** scans PATH for claude/codex/gemini/ollama, probes localhost:11434; shows found list with checkboxes.
- **Add API providers:** endpoint + key (stored to keychain, FR-2) -> auth test -> model discovery.
- **Quota setup:** for each provider pick source (auto / manual budget) + plan limits.
- **Default rules:** proposed from capabilities (e.g. tool-runner -> ollama local); user confirms in the rule editor.
- Done -> Mission Control. maestro config doctor summary shown.

## 7.2 New project

- Ctrl+N -> wizard: name, dir, stack, git init.
- **Interview** (interrogator role, FR-6): batched forms; answers stream into PRD.json.
- Maestro drafts **PRD.md** -> user reviews in markdown tab -> approve -> PRD.json compiled.
- **AGENTS.md + skill suggestions** generated; user checks which skills to install.
- **Task DAG proposal** shown in the project's tree; user edits/reorders -> **Run**.
- Sessions spawn per rule routing; dashboard shows them; questions may reappear mid-flight in the app bar ([?] badge).

## 7.3 Quota-exceeded event (no user action needed)

- Provider returns 429 / parsed quota hits threshold -> rule re-evaluates (FR-3.3).
- Session migrates via structured snapshot (D5); transcript gains a — migrated kimi-k2.7 -> glm-5.2 — marker.
- App bar [!] badge + notification; Activity tab records the decision and its reason.

## 7.4 Steering a running agent

Open session window -> type in steer box -> message injected into agent loop. ■ Stop pauses with snapshot; resume continues with context intact.

## 8. Keybindings (defaults, remappable)

| Key        | Action                                                  |
|------------|---------------------------------------------------------|
| Ctrl+Q     | Quick Chat                                              |
| Ctrl+N     | New Project                                             |
| F10        | Settings                                                |
| F1..F6     | Mission Control tabs (Dashboard...Activity)             |
| Ctrl+Enter | Send (chat/steer)                                       |
| Ctrl+P     | Command palette (jump to project/provider/rule/session) |
| Ctrl+D     | Dry-run rule under cursor                               |
| Ctrl+T     | Tile all satellite windows                              |
| Esc        | Close dialog / blur input                               |
| ?          | Shortcuts overlay                                       |

## 9. AppCUI Control Mapping

| Need                               | AppCUI control                                      |
|------------------------------------|-----------------------------------------------------|
| Tabs everywhere                    | Tab                                                 |
| Sessions/providers/projects tables | ListView (columns, sort)                            |
| Task DAG                           | TreeView + custom glyphs                            |
| Quota gauges                       | ProgressBar + GraphView (burn sparklines)           |
| PRD view                           | Markdown viewer                                     |
| Chat transcript                    | RichTextField (read-only, appended)                 |
| Steer/chat input                   | TextField (multiline) + Button                      |
| Rule builder                       | ComboBox, NumericField, ListBox + up/down           |
| Settings sections                  | Accordion / ListBox + panel swap                    |
| Project window panes               | Splitter                                            |
| Global status                      | AppBar segments                                     |
| Wizards/onboarding                 | modal Windows + Buttons, CheckBox, RadioButton      |
| Questions forms                    | generated RadioButton/CheckBox/TextField/DatePicker |
| Toasts/alerts                      | notification dialogs + app bar badges               |
| Theming                            | built-in themes, true-color                         |

## 10. Window Inventory

| Window              | Kind             | Opened from                            |
|---------------------|------------------|----------------------------------------|
| Mission Control     | main (maximized) | startup                                |
| Onboarding wizard   | modal            | first run / Tools                      |
| Project window      | satellite        | Projects tab, maestro project open     |
| Session/Chat window | satellite        | dashboard/chat tab, Quick Chat promote |
| Quick Chat          | satellite        | Ctrl+Q                                 |
| Provider window     | modal-ish        | Providers tab                          |
| Provider wizard     | modal            | Providers -> Add                       |
| Rule editor         | tab panel        | Rules tab                              |
| Interrogation form  | panel/modal      | Questions tab, app bar [?]             |
| Settings            | modal            | F10                                    |
| Transcript viewer   | satellite        | session -> [log] full                  |
| File/picker dialogs | stock AppCUI     | wizards                                |